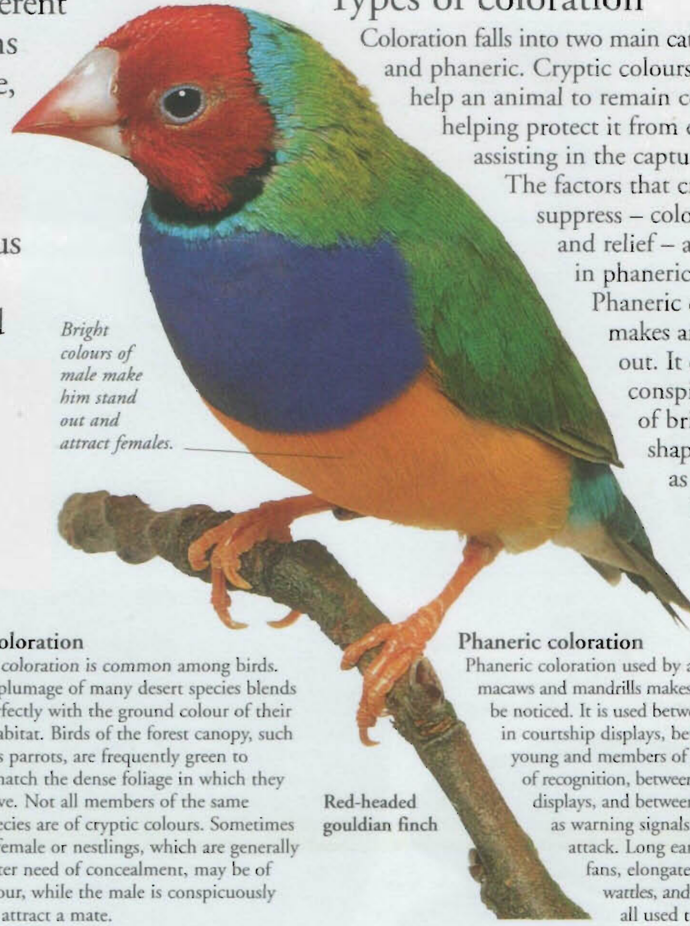
CAMOUFLAGE AND COLOUR

ANIMALS HAVE EVOLVED different colours, shapes, and patterns that help them survive. Some, such as birds-of-paradise, are brightly coloured to attract a mate; others, such as the fire salamander, use colour to advertise that they are poisonous to eat. Animals, such as lapwings and polar bears, are camouflaged – coloured or patterned – in such a way that they blend with their surroundings. Camouflage helps animals to hide from predators, but it can also help predators to creep up on their prey.

Types of coloration

Coloration falls into two main categories: cryptic and phaneric. Cryptic colours and patterns help an animal to remain concealed, thus helping protect it from enemies, or assisting in the capture of its prey. The factors that cryptic species suppress – colour, movement, and relief – are exaggerated in phaneric species. Phaneric coloration makes an animal stand out. It can include the conspicuous display of brilliant colours, shapes, and actions, as demonstrated by birds-of-paradise.



Bright colours of male make him stand out and attract females.

Red-headed gouldian finch

Newly hatched lapwings match colour of straw.

Young lapwings in nest

Cryptic coloration

Cryptic coloration is common among birds. The plumage of many desert species blends perfectly with the ground colour of their habitat. Birds of the forest canopy, such as parrots, are frequently green to match the dense foliage in which they live. Not all members of the same species are of cryptic colours. Sometimes the female or nestlings, which are generally in greater need of concealment, may be of cryptic colour, while the male is conspicuously coloured to attract a mate.

Phaneric coloration

Phaneric coloration used by animals such as macaws and mandrills makes them stand out and be noticed. It is used between male and female in courtship displays, between parent and young and members of a group for purposes of recognition, between rival males in threat displays, and between predators and prey as warning signals, bluff, or to deflect attack. Long ear- and head-plumes, fans, elongated tail feathers, wattles, and inflatable air sacs are all used to attract attention.

Camouflage

For concealment to be effective, the colour and pattern of an animal's coat or skin must relate closely to its background. A bird's colour often harmonizes with its nesting requirements; some ground-nesting birds choose a nest site with surroundings of similar colour to their eggs as an aid to concealment. Colour and posture can be a highly effective form of camouflage. The many types of concealment include disruptive coloration, disguise, and immobility.

Disruptive coloration

Irregular patches of contrasting colours and tones of an animal's coat divert attention away from the shape of the animal, making it harder to recognize. Tigers and giraffes show disruptive coloration.

Tiger camouflaged in long grass

Giant spiny stick insect

Disguise

Cryptic coloration aims to disguise rather than conceal. The combination of colour, form, and posture can produce an almost exact replica of a commonplace object associated with the habitat. Stick insects, for example, resemble small twigs, while nightjars, when lying down, look like stones or wood fragments.

Mimicry

Mimicry is an extreme form of concealment. It occurs when a relatively defenceless or edible species looks like an aggressive or dangerous species. The mimic not only takes on the appearance of the object it is mimicking, but also adopts its behaviour, assuming characteristics that are completely alien to it. For example, harmless milk snakes resemble poisonous coral snakes so that other animals will not attack them. The monarch, a poisonous butterfly, is mimicked by a non-poisonous species, *Hypolimnus*, which is indistinguishable from it.

Coral snake

Milk snake

Milk snakes have stripes of the same colour as coral snakes, but in a different order.

Reedbuck

Immobility

Effective camouflage is possible only if an animal remains still. Many animals react to danger by freezing. For example, if confronted with danger, reedbuck crouch down with their necks outstretched, and by remaining motionless, become hard to distinguish from their surroundings. Some birds, particularly ground-nesting birds such as nightjars, squat down to reduce the shadow they make.

Assassin bug

Many species of assassin bugs resemble the insects on which they feed. This enables them to get close to their prey without being detected, before seizing it and injecting a toxic fluid. One species of assassin bug, *Salyavata variegata*, lives in termite nests. It camouflages itself by covering its body in debris, including the bodies of termites, and then enters the nest, unnoticed, to feed on the inhabitants.

Assassin bug covered in debris by termites' nest

Termite